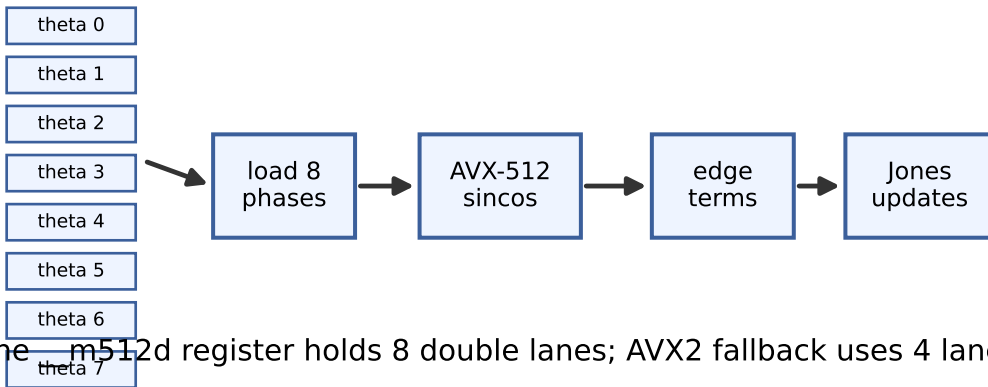




one m512d register holds 8 double lanes; AVX2 fallback uses 4 lanes

the code batches trigonometric phases and dot products across detector directions or sizes